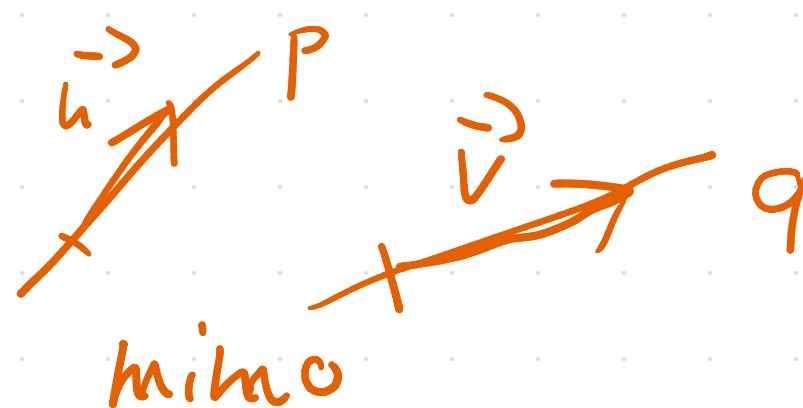
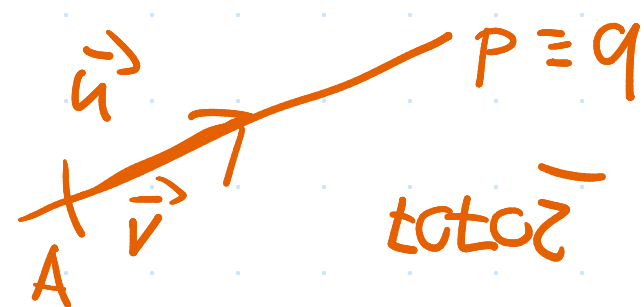
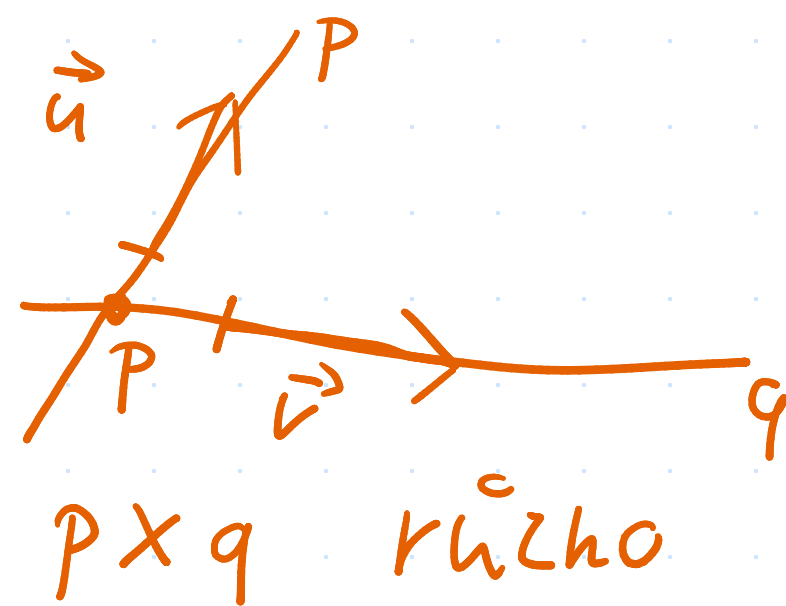
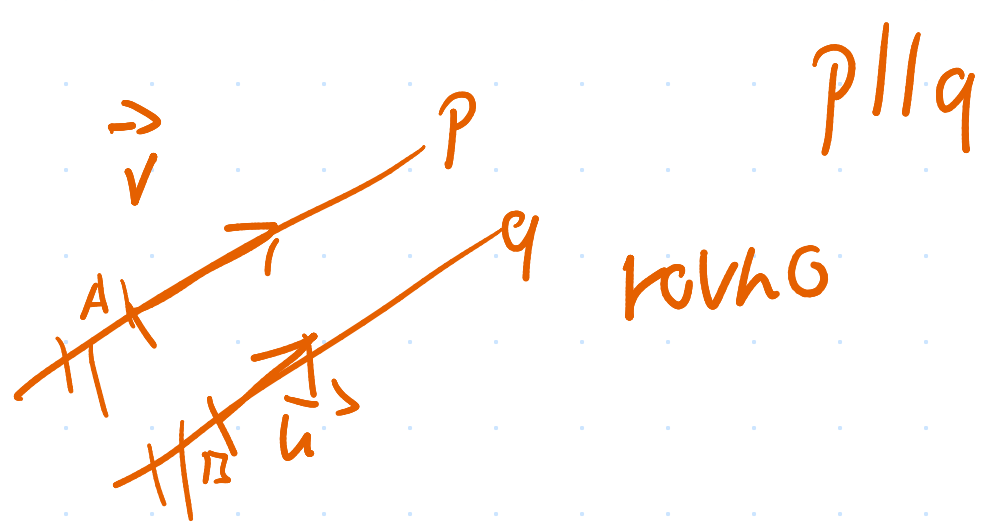


Vzájemná poloha přímek v prostoru



$$1) \quad a: \begin{cases} x = -1 - 8s \\ y = 16 + 4s \\ z = -2s \end{cases} \quad p: \begin{cases} x = 3 + 4t \\ y = 14 - 2t \\ z = 1 + t \end{cases}$$

$$\left. \begin{aligned} -8 = 4t &\Rightarrow t = -2 \\ 4 = -2t &\Rightarrow t = -2 \\ -2 = t & \end{aligned} \right\} -2 = -2 = -2 \quad \begin{cases} \text{rovno} \\ \text{totož} \end{cases}$$

$$A \in p(?) \quad \left. \begin{aligned} -1 &= 3 + t \cdot 4 \Rightarrow t = -1 \\ 16 &= 14 - 2t \Rightarrow t = -1 \\ 0 &= 1 + t \Rightarrow t = -1 \end{aligned} \right\} -1 = -1 = -1 \Rightarrow A \in p$$

$\Downarrow$
 $p = a$

$$2) \quad c: \begin{cases} x = 7 + 4s \\ y = 4s \\ z = 2s \end{cases} \quad p: \begin{cases} 3 + 4t = x \\ 14 - 2t = y \\ 1 + t = z \end{cases}$$

$$\left. \begin{aligned} 4 &= 4t \Rightarrow t = 1 \\ 4 &= -2t \Rightarrow \\ 2 &= t \end{aligned} \right\} 1 \neq 2 \quad \begin{cases} \text{různ} \\ \text{mimo} \end{cases}$$

$$7 + 4s = 3 + 4t$$

$$4s = 14 - 2t$$

$$2s = 1 + t \Rightarrow t = 2s - 1$$

$$7 + 4s = 3 + 8s - 4$$

$$4s = 14 - 4s + 2$$

$$8 = 4s \Rightarrow s = 2$$

$$8s = 16 \Rightarrow s = 2$$

$$\left. \begin{array}{l} 8 = 4s \Rightarrow s = 2 \\ 8s = 16 \Rightarrow s = 2 \end{array} \right\} 2 = 2 \Rightarrow P \times q$$

růžno

$$x = 7 + 4 \cdot 2 = 15$$

$$y = 8$$

$$z = 4$$

$$P = [15; 8; 4]$$

vzájemná poloha přímky a roviny

$$3) p: x = t$$

$$y = t$$

$$z = 1 + 3t$$

$$\vec{s}_p = (1; 1; 3)$$

$$A = [0; 0; 1]$$

$$\rho: 2x + y - z + 1 = 0 \quad \vec{n}_\rho = (2; 1; -1)$$

$$p \parallel \rho (?): \vec{n}_\rho \perp \vec{s}_p \Rightarrow \vec{n}_\rho \cdot \vec{s}_p = 0$$

$$(2; 1; -1) \cdot (1; 1; 3) = 0$$

$$2 + 1 - 3 = 0$$

$$0 = 0$$

$$2x + y - z + 1 = 0$$

$$2 \cdot 0 + 0 - 1 + 1 = 0$$

$$0 = 0 \Rightarrow A \in \rho \Rightarrow p \subset \rho$$

vzdálenost bodu od přímky

vzorec neexistuje, jen postup

$$1) \rho: ax + by + cz + d = 0$$

$$\vec{n}_\rho = \vec{s}_\rho \quad \wedge \quad M \in \rho$$

$$2) P \in \rho \cap \rho$$

$$3) V = |MP| = \sqrt{(x^2 + y^2) \dots}$$

$$4) M = [0; 1; 4]$$

$$\rho: x = 2 + t$$

$$y = 3$$

$$z = 2 - t$$

$$\vec{s}_\rho = (1; 0; -1) = \vec{n}_\rho$$

$$ax + by + cz + d = 0$$

$$x - z + d = 0$$

$$4 + d = 0$$

$$d = -4$$

$$\rho: x - z - 4 = 0$$

$$2 + t - (2 - t) - 4 = 0$$

$$2 + t - 2 + t - 4 = 0$$

$$-4 + 2t = 0$$

$$t = 2$$

$$x = 2 + 2 = 4$$

$$z = 0$$

$$P = [4; 3; 0]$$

$$V = |\vec{MP}| = \sqrt{(4-0)^2 + (3-1)^2 + (0-4)^2} =$$

$$= \sqrt{16 + 4 + 16} = \sqrt{36} = \underline{\underline{6}}$$

přesečnice rovin

$$5) \left. \begin{array}{l} \rho: 2x - y - z - 1 = 0 \\ \sigma: x + y + 2z - 3 = 0 \end{array} \right\} (+)$$

$$\underline{3x + z - 4 = 0}$$

$$3x + z - 4 = 0$$

$$z = 4 - 3x$$

$$z = 4 - 3t$$

$$\boxed{x = t}$$

$$2t - y - 4 + 3t - 1 = 0$$

$$-y + 5t - 5 = 0$$

$$5t - 5 = y$$

závěr

$$x = t$$

$$y = 5t - 5$$

$$z = 4 - 3t$$